# Discrete Mathematics

1001

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# 第一章　Logic and Proofs

**The Only Truth is the Truth Table.**

## 1.1　Basic

**Proposition**: A declarative sentence that is either True or False, but not both.
not declarative: $x + 1 = 3$
not T or F: Paradox. e.g. This sentence is false.
allow its truth value be undetermined: He will go to Paris.

**Proposition Varible**: $p, q, r$, etc.

**Atomic Proposition**: the simplest proposition.

Given proposition $p \to q$:

| | |
|---|---|
| **Converse** 逆命题 | $q \to p$ |
| **Inverse** 否命题 | $\neg p \to \neg q$ |
| **Contrapositive** 逆否命题 | $\neg q \to \neg p$ |

$p$: **Hypothesis** / **Antecedent** / **Premise**
$q$: **Conclusion** / **Consequence**

**Tautology**: always True;
**Contradiction**: always False;
**Contingency**: sometimes True, sometimes False;
**Satisfiable**: 存在赋值方式使为真, Tautology + Contingency

Precedence: $\forall/\exists > \neg > \wedge/\vee > \to / \leftrightarrow$

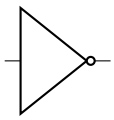
**Theorem**: a somewhat important statement.
**Proposition** / Facts / Results: less important.
**Axiom** 公理 / **Postulate** 假设: can be used as premises.
**Lemma**: a less important theorem that is helpful for other results.
**Corollary** 推论: a theorem that can be established directly from a theorem.
**Conjecture** 猜想: guess to be true.

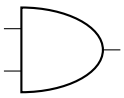
# 1.2 Logical Operators

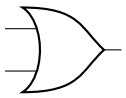
## 1.2.1 Defines

| | | | | |
|---|---|---|---|---|
| Negation 否定 | $\neg x$ / $!x$ / $\sim x$ | not | | $1-x$ |
| Conjunction 合取 | $x \wedge y$ | and | | $xy$ |
| Disjunction 析取 | $x \vee y$ | or | | $x+y-xy$ |
| Exclusive Or | $x \oplus y$ | xor | | |
| Conditional / Implication 蕴涵 | $x \to y$ | if, then | $\neg x \vee y$ | $1-x+xy$ |
| Biconditional / Equivalence 等价 | $x \leftrightarrow y$ | iff | $\neg(x \oplus y)$ | |

x only if y: $x \to y$
x unless y: $y \to \neg x$
x, y both occur: $x \wedge y$, no causality.
avoid using *any*, 因为自然语言下同时有任意一个/某一个的含义。


Inverter


And Gate


Or Gate

## 1.2.2 Rules

Logical Equivalence:

- Identity: $x \wedge \mathbf{T} \equiv x$, $x \vee \mathbf{F} \equiv x$
- Domination: $x \wedge \mathbf{F} \equiv \mathbf{F}$, $x \vee \mathbf{T} \equiv \mathbf{T}$
- Idempotent: $x \wedge / \vee x \equiv x$
- Double Negation: $\neg(\neg x) \equiv x$
- Communtative & Associative Laws: $\wedge/\vee$ 分别满足交换 & 结合律
- Distribution: $\wedge/\vee$ 分别对 $\vee/\wedge$ 分配
  $$\begin{cases} x \vee (y \wedge z) & \equiv (x \vee y) \wedge (x \vee z) \\ x \wedge (y \vee z) & \equiv (x \wedge y) \vee (x \wedge z) \end{cases}$$
- De Morgan's Laws: flip operator
  $$\begin{cases} \neg(x \wedge y) & \equiv \neg x \vee \neg y \\ \neg(x \vee y) & \equiv \neg x \wedge \neg y \end{cases}$$
- Absorption Laws:
  $$\begin{cases} x \vee (x \wedge y) & \equiv x \\ x \wedge (x \vee y) & \equiv x \end{cases}$$

### 1.2.3 Normal Forms

Simple Conjunctive/Disjunctive Formula: only consists of $\wedge/\vee$ and $\neg$

Conjunctive/Disjunctive Normal Form (CNF / DNF): $\bigwedge(\vee)$ / $\bigvee(\wedge)$
主合取/析取范式：外层为合取/析取。CNF / DNF 必存在，构造性证明，略。

CNF / DNF 不唯一，故引出 **Principle/Major C/D NF** (P*NF/M*NF)（主合/析取范式），强制约定每个基本项包含所有 $n$ 个变元。 P*NF 是唯一的。

极小项：$\bigwedge_{i=1}^{n}$, $m_0 \sim m_{2^n-1}$；
极大项：$\bigvee_{i=1}^{n}$, $M_0 \sim M_{2^n-1}$；
编号：from left to right, $i$-th bit $= [p_i]$。
e.g. $(p_1 \wedge \neg p_2 \wedge \neg p_3) = m_{100} = m4$; $(\neg p_1 \vee p_2) = M_{01} = M_1$

Construct P*NF base on *NF: for each simple formula with less than $n$ varibles, adjust, and merge the same formulas
a. $A \equiv (A \wedge p_k) \vee (A \wedge \neg p_k)$
b. $A \equiv (A \vee p_k) \wedge (A \vee \neg p_k)$

转换 PCNF/PDNF，其中 $\sim$ 为按位取反：$\bigwedge\limits_{t\in S} M_t = \bigvee\limits_{t\in U\setminus S} m_{\sim t}$； $\bigvee\limits_{t\in S} m_t = \bigwedge\limits_{t\in U\setminus S} M_{\sim t}$
证明：PDNF $\to$ PCNF：每个 $t \in S$ 表示一种成真赋值，则每个 $t \in U\setminus S$ 表示一种成假赋值。为使结果为真，则每种成假赋值均需有一项不成立，即补集的按位取反。易知该操作可逆，则 PCNF $\to$ PDNF 亦得。

## 1.3 Predicates and Quantifiers

### 1.3.1 Defines

**Predicate** 谓词: refer to a property, propositonal function $P$, $P(x_1, \cdots x_n)$

**Quantifier** 量词: the extent to which $P$ is true over a range of elements
a. universal quantifier 全称量词: $\forall$
b. existential quantifier 存在量词: $\exists$
c. uniqueness quantifier 唯一量词: $\exists!/\exists_1$

**Domain**: the domain of $x$; the domain consists of $a, b, c, \cdots$
**Scope**: the scope of $\forall/\exists$, etc.

Bound Occurance 约束出现: $x$ is restricted by quantifiers.
Free Occurance 自由出现: $x$ is not restricted by any quantifiers.
Closed Form 闭式: no free varibles, all varibles is in some scope.

### 1.3.2 Rules

- Negation:
  a. $\neg\forall \equiv \exists\neg$
  b. $\neg\exists \equiv \forall\neg$

- Distribution: turn $\rightarrow$ / $\leftrightarrow$ to $\wedge/\vee$ first
  a. under the same restriction: only the following two are correct
  - $\forall x(A(x) \wedge B(x)) \equiv \forall x A(x) \wedge \forall x B(x)$
  - $\exists x(A(x) \vee B(x)) \equiv \exists x A(x) \vee \exists x B(x)$

  b. under different restriction: $\forall/\exists x(A(x) \wedge / \vee B) \equiv \forall/\exists x A(x) \wedge / \vee B$

### 1.3.3 Prenex Normal Form

前束范式: all quantifier are at the beginning, and their scopes last to the end.
e.g. $\forall x \exists y \forall z(P(x,y) \vee Q(x,y,z) \vee \neg R(z))$

PNF 必存在，构造性证明，略。
PNF 不唯一。

## 1.4 Rules of Inference

a **valid argument**: ($\bigwedge$ premises) $\rightarrow$ conclusion

**fallacy**: 谬误

### 1.4.1 Rules

| | | |
|---|---|---|
| Modus Ponens | 肯定前件 | $\begin{cases} x \\ x \rightarrow y \end{cases} \implies y$ |
| Modus Tollens | 否定后件 | $\begin{cases} \neg y \\ x \rightarrow y \end{cases} \implies \neg x$ |
| Hypothetical Syllogism | 假言三段论 | $\begin{cases} x \rightarrow y \\ y \rightarrow z \end{cases} \implies x \rightarrow z$ |
| Disjunctive Syllogism | 析取三段论 | $\begin{cases} x \vee y \\ \neg x \end{cases} \implies y$ |
| Addition | | $x \implies (x \vee y)$ |
| Simplification | | $(x \wedge y) \implies x$ |
| Conjunction | | $\begin{cases} x \\ y \end{cases} \implies (x \wedge y)$ |
| Resolution | 消解 | $\begin{cases} (x \vee y) \\ (\neg x \vee z) \end{cases} \implies (y \vee z)$ |

| | | |
|---|---|---|
| Constructive Dilemma | | $\begin{cases}(x \to y) \land (u \to v) \\ x \lor u\end{cases} \implies (y \lor v)$ |
| Destructive Dilemma | | $\begin{cases}(x \to y) \land (u \to v) \\ (\neg y \lor \neg v)\end{cases} \implies (\neg x \lor \neg u)$ |
| Universal Instantiation (UI) | | $\forall x P(x) \implies P(u)$ |
| Existential Instantiation (EI) | | $\exists x P(x) \implies P(u)$ for some u |
| Universal Generalization (UG) | | $P(u)$ for an arbitrary $u \implies \forall x P(x)$ |
| Existential Generalization (EG) | | $P(u)$ for some $u \implies \exists x P(x)$ |
| Universal Modus Ponens | | $\begin{cases}\forall x(P(x) \to Q(x)) \\ P(u)\end{cases} \implies Q(u)$ |
| Universal Modus Tollens | | $\begin{cases}\forall x(P(x) \to Q(x)) \\ \neg Q(u)\end{cases} \implies \neg P(u)$ |

prep.: using / from

e.g.

| | | |
|---|---|---|
| 1. | $\neg t$ | premise 3. |
| 2. | $r \to t$ | premise 2. |
| 3. | $\neg r$ | modus tollens using 1, 2 |
| 4. | $\neg r \lor \neg s$ | addition from 3 |
| 5. | $\neg(r \lor s)$ | De Morgan's |

### 1.4.2 Fallacies

- fallacy of affirming the conclusion:
  $\begin{cases}x \to y \\ y\end{cases} \not\Rightarrow x$
- fallacy of denying the hypothesis:
  $\begin{cases}x \to y \\ \neg x\end{cases} \not\Rightarrow \neg y$

## 1.5 Proof Methods

Direct Proof.

Conditional Proof (CP) 附加前提证明: to prove $A \implies (B \to C)$, use $B$ as a premise. try to prove $(A \land B) \to C$.

Proof by Contraposition: use $\neg q \to \neg p$ to prove $p \to q$.

Proof by Contradiction 反证

Proof by Counterexample 找反例: use $\neg P(u)$ to prove $\neg\forall x P(x)$.

Existence Proof:
a. Constructive: find a witness, that $P(u)$ is true.
b. Nonconstructive

Exhaustive Proof / Proof by Exhaustion 穷举
Proof by Cases 分讨

Tips:

- Vacuous Proof: if $p$ is false, then $p \to q$ is immediately true.
- Trivial Proof: if $q$ is true, then $p \to q$ is always true.
- proof of equivalence:
  a. to prove $p \leftrightarrow q$, proving $(p \to q) \wedge (q \to p)$.
  b. to prove $p_1 \leftrightarrow p_2 \leftrightarrow \cdots \leftrightarrow p_n$, proving $\bigwedge(p_i \to p_{i \bmod n+1})$.
- **WLOG**: without loss of generality.

# 第二章　Relations

Can't use emptyset for Cartesian Product.

## 2.1 Definitions

A binary relation $R$ from $A$ to $B$ is a subset of $A \times B$: $R \subset A \times B$

If $(a, b) \in R$, then $a$ is related to $b$. The **order** matters. Denoted as $aRb$ or $(a, b) \in R$. Otherwise, $a \not R b$ or $(a, b) \notin R$.

The length of tuple is called degree. e.g. The degree of $A \times \cdots \times A$ is $n$.
$A_i$ is called the domains of $R$.

### 2.1.1 Representation

**Order!**

- **directed** bipartite graph
- **directed** graph for $R \subset A \times A$
- 01-matrix: $R : A \to B$, $M_{i,j} = [(a_i, b_j) \in R]$
  $M_{R \cap S} = M_R \wedge M_S$; $M_{R \cup S} = M_R \vee M_S$
  $M_{S \circ R} = M_R \odot M_S$ a.k.a. $M_{i,j} = \vee(M_{S_{i,k}} \wedge M_{R_{k,j}})$ **矩阵顺序不一样!**
  $M_{R^n} = M_R^n$

### 2.1.2 Special Relations

- 空关系 $\emptyset$
- 恒等关系 $I_A = \{(a, a) : \forall a\}$
- 全域关系 $U_A = \{(a, b) : \forall a \forall b\}$

### 2.1.3 Properties

- 自反性 Reflexive: $\forall x \in A, xRx$
  Each node has a self-loop.
  All elements on the main diagonal are 1.

- 反自反性 Irreflexive: $\forall x \in A, (x, x) \notin R$
  Irreflexive is **not** the negation of reflexive !
  $R$ can be neither reflexive nor irreflexive. $R$ is both reflexive and irreflexive iff $A = \emptyset$ (however, Cartesian Product asks $A \neq \emptyset$ in $A \times A$, so?).
  Each node must not have a self-loop.
  All elements on the main diagonal are 0.

- 对称性 Symmetric: $R = R^{-1}$ (a.k.a. $\forall x, y \in A\ (x, y) \in R \iff (y, x) \in R$)
  （等价地）All edges are bi-directed.
  All are same across the main diagonal.

- 反对称性 Antisymmetric: $(x, y) \in R \wedge (y, x) \in R \implies x = y$
  Antisymmetric is **not** "not symmetric" !
  For any two different vertices, there are at most one directed edge link them.
  All 1's are across from 0's.

- 非对称性 Asymmetric: $(x, y) \in R \implies (y, x) \notin R$
  Completely no symmetric. Asymmetric $\iff$ Antisymmetric $\wedge$ Irreflexive.
  No self-loops, and for any two different vertices, there are at most one directed edge link them.
  All elements on the main diagonal are 0, and all 1's are across from 0's.

- 传递性 Transitive: $(x, y), (y, z) \in R \implies (x, z) \in R$

### 2.1.4 Operations

Inverse Relation: $R^{-1} := \{(b, a) : (a, b) \in R\}$.
e.g. $R = \{(a, b) : a < b\}$, then $R^{-1} = \{(a, b) : a > b\}$.
Note that if $R = \{(a, b) : p(a, b)\}$, $R^{-1}$ is **not** $\{(a, b) : \neg p(a, b)\}$ !

Properties of Inverse Relation:
a. $(R^{-1})^{-1} = R$
b. $(R\ \cap / \cup / -\ S)^{-1} = R^{-1}\ \cap / \cup / -\ S^{-1}$
c. $R \subset S \iff R^{-1} \subset S^{-1}$
d. $(R \circ S)^{-1} = S^{-1} \circ R^{-1}$

Composition: $S \circ R : A \overset{R}{\to} B \overset{S}{\to} C$

Power:
a. $R^0 := I$
b. $R^n := R^{n-1} \circ R$
c. $R^{-n} := (R^{-1})^n$

**Thm**. $R$ is transitive iff $R^n \subset R$ for all $n \in \mathbb{N}_+$

**Thm**. Composition of relations satisfies the associative law.

## 2.2 Closures of Relations

### 2.2.1 Defination

For any property P, the "P closure" of $R$, if it exists, is defined as the "smallest" subset of $A \times A$ containing $R$ with P.

- contains $R$
- has property $P$
- "smallest": for every $R \subset T$ s.t. $T$ has property P, $S \subset T$.
  因此，闭包是合法集合，且是所有合法 $T$ 的交集。若把集合的包含关系看作合法集之间的偏序关系，则存在闭包说明存在**唯一**的极小元，这对 P 本身有所限制。

closures:

- reflexive closure: $r(R) = R \cup I_A$
- symmetric closure: $s(R) = R \cup R^{-1}$
- transitive closure: $t(R) = R^* = \bigcup\limits_{n \in \mathbb{N}_+} R^n$

### 2.2.2 Connectivity Relation

a.k.a. transitive closure

$(a, b) \in R^n$: there exists a path from $a$ to $b$ with length $n$

$|A| = n$, if there is a path $a \rightsquigarrow b$, then there must exist a path with length $\leqslant n$. Therefore, $t(R) = R \cup \cdots \cup R^n$

- naïve algo: brute force $O(n^4)$
- Warshall's: $w_{i,j} \leftarrow w_{i,k} \wedge w_{k,j}$, $O(n^3)$
  interior vertices: exclude $a, b$
  init: $W = M_R$

## 2.3 Equivalence Relations

## 2.4 Partial Orderings